

API SPECIFICATION DOCUMENT

PART-4: API GOVERNANCE, MONITORING & OPERATIONS

School ERP

Spring Boot 3 + Java 21 | PostgreSQL | Redis

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Preamble

This document is Part-4, the final part, of the API Specification Document (ASD) v1.0 for the School ERP System, continuing directly from Parts 1-3. It defines API lifecycle, deprecation, documentation, monitoring, performance, testing, change-management, release, and governance standards — the operational and organizational counterpart to the design/security/processing standards already fixed in ASD Parts 1-3.

This document does not regenerate RGD v2.0, any prior Appendix, any UI/UX Screen Specification, any MDB Part, any DDD Part, or any prior ASD Part. It contains no source code, no controller code, no JSON, no API endpoint list, no OpenAPI/Swagger specification content, no sequence diagrams, and no examples — every standard is stated as a rule only.

31. API Lifecycle Management

Fixes the operational management of the API Lifecycle already conceptually defined in Appendix-K Section 23.

- Every API version (ASD Part-1 Section 7) progresses through Design, Active, Deprecated, and Retired stages in strict sequence — a version is never reintroduced as Active after being Deprecated, and never skips directly from Active to Retired without a Deprecated period (Section 32).
- An API Group's stage is tracked centrally, not inferred implicitly from whether client code still calls it — the API governance owner (Section 39) maintains the authoritative record of every version's current stage.
- Lifecycle stage transitions follow the same environment-promotion discipline already fixed in MDB Part-4 Section 31/35 — a version does not become Active in Production before having been Active in Staging against production-representative traffic patterns.
- No API Group in the current specification (Appendix-K Section 18) is in the Deprecated or Retired stage as of this document's approval, consistent with the system's initial-release status already noted in Appendix-K Section 23.

32. API Deprecation Standards

Fixes the concrete deprecation process realizing Appendix-K Section 23's lifecycle model and Section 24's Backward Compatibility rules.

- A version is marked Deprecated only after its replacement is fully Active and has passed the same Testing/UAT/Staging gates (Appendix-M Sections 20-21) already required for any new release.
- Every Deprecated-version response carries a deprecation indicator (ASD Part-1 Section 10's metadata element), giving the calling client (React frontend, or any future third-party integration) a machine-readable signal ahead of retirement, not solely a documentation notice.
- The deprecation notice period (time between marking Deprecated and Retired) remains Client/Product Decision Required, but is never zero — a version is never retired in the same release cycle it is deprecated.
- Retirement (removal of a Deprecated version) follows the same Change Management approval already fixed in Section 37 this document — retirement is a governed decision, not an automatic cleanup action.

33. API Documentation Standards

Fixes the requirement that a machine-readable API documentation artifact exists and stays current, without this ASD itself containing that artifact's content, per this document's no-OpenAPI/no-Swagger scope.

- A concrete, OpenAPI-class specification (or equivalent machine-readable interface description) is generated and maintained as a build artifact (MDB Part-1 Section 10), separate from this ASD — this ASD fixes the conventions that artifact must conform to (Parts 1-3), not the artifact's content.
- The documentation artifact is regenerated as part of the same build pipeline already fixed in MDB Part-4 Section 34-35, so it can never drift out of sync with the deployed API — a manually-maintained, independently-updated documentation artifact is not permitted.
- Every API Group's documentation entry references its governing FR/BR/Screen ID (ASD Part-1 Section 2), preserving the traceability chain from RGD v2.0 through to the published interface description.
- Internal (module-to-module, Administration package) interfaces are documented separately from externally-consumable API Groups, consistent with the shared-service distinction already fixed in MDB Part-1 Section 8 — internal interfaces are not published in the same client-facing documentation artifact.

34. API Monitoring Standards

Fixes API-specific monitoring signals extending the general Monitoring & Health Standards already fixed in MDB Part-4 Section 37 and DDD Part-4 Section 36.

Metric	API-Specific Signal
Per-Operation Latency	Response time tracked per API Group and per HTTP method (ASD Part-1 Section 6), not only as a system-wide average, since different operations carry different targets (Appendix-F NFR-PERF series)
Per-Operation Error Rate	Error category breakdown (ASD Part-3 Section 28) tracked per API Group, distinguishing Validation/Business Rule/Authorization/System errors rather than a single combined error count
Rate Limit Activity	Redis-backed rate-limit rejection counts (ASD Part-2 Section 17) tracked separately from general error rate, to distinguish abuse/overload from genuine defects
Authentication/Authorization Failures	Tracked per ASD Part-2 Sections 11-12, feeding the same account-lockout and unusual-access-pattern alerting already fixed in Appendix-F NFR-MON-001
Version Usage	Traffic volume per API version (Section 31), informing when a Deprecated version's usage has dropped enough to safely proceed toward Retirement (Section 32)

35. API Performance Standards

Consolidates the API-level Performance targets already fixed across Appendix-F, ASD Part-1 Section 9, and DDD Part-3 Section 27 into a single operational reference.

Operation Class	Target (Already Fixed)
General API Response (95th percentile)	≤500 ms (NFR-PERF-011)
Search Operations (Section 26, ASD Part-3)	≤2 seconds (NFR-PERF-004, NFR-SRCH-001)
Payment Recording	≤5 seconds gateway-confirmation-to-response (NFR-PERF-007)
Large Report/Export (Section 27, ASD Part-3)	Background job beyond the row-count threshold (NFR-RPTP-001)

These targets are not re-derived here — they are the same targets already fixed in Appendix-F, restated as the single API-level performance reference so an API Architecture Team member does not need to cross-reference the full NFR catalogue for routine performance-review purposes.

36. API Testing Standards

Fixes API-layer application of the Test Levels already established in Appendix-M Section 8, without redefining those levels.

- Contract-level testing (verifying an API Group's request/response shape matches this ASD's standards, Parts 1-3) is performed as part of Integration Testing (Appendix-M Section 8), before an API change is eligible for promotion beyond the Testing environment.
- Every Business Rule already required to have Unit Testing coverage (Appendix-M Section 9, MDB Part-3 Section 29) is additionally verified at the API layer during Integration Testing, confirming the rule's violation surfaces the correct response category (ASD Part-3 Section 22), not only that the Service-layer logic itself is correct.
- Regression Testing (Appendix-M Section 8) for any API change includes verification that Backward Compatibility (ASD Part-1 Section 7) has not been broken for any still-Active or Deprecated version.
- Non-Functional Testing for API Performance (Section 35 this document) and Security (ASD Part-2 Section 20) both occur in Staging against production-representative load, consistent with Appendix-M Section 10's environment requirement.

37. API Change Management

Fixes the approval discipline for any change to a published API Group, extending the general Change Management already fixed in MDB Part-4 Section 39 and DDD Part-4 Section 39.

- Every proposed API change is classified as additive/non-breaking or breaking (ASD Part-1 Section 7's Backward Compatibility test) before implementation begins — this classification determines whether a version increment (Section 31) is required.
- A breaking change requires sign-off from the API Architecture Team (Section 39) in addition to the standard Code Review Checklist (MDB Part-3 Section 30) and the API-specific checklists already fixed in ASD Part-2 Section 20 and ASD Part-3 Section 30.
- Frontend (React + TypeScript) and Backend (Spring Boot 3) changes to the same API Group are coordinated as a single change record, even when implemented in separate commits/pull requests, preventing a deployed frontend from ever calling a not-yet-deployed backend contract or vice versa.
- An emergency API change (responding to an active Production incident, MDB Part-4 Section 39) follows the same retroactive-review requirement already fixed there — an emergency change is never left permanently outside this governance record.

38. API Release Standards

Fixes API-specific release coordination extending the Build & Release Standards and Deployment Strategy already fixed in MDB Part-4 Sections 34 and 36.

- An API release is defined as a specific, tagged combination of backend API implementation and, where applicable, its corresponding frontend consumer — matching the release definition already fixed in MDB Part-4 Section 34.
- Non-breaking API additions (ASD Part-1 Section 7) may release independently of a coordinated frontend release, since existing clients remain unaffected by an additive change — a breaking change (Section 37) requires coordinated release of both frontend and backend.
- API release notes reference the specific FR/BR/Screen IDs affected (ASD Part-1 Section 2), extending the traceability discipline already fixed throughout this documentation set into the release-communication process itself.
- Production API releases occur within the same planned maintenance window discipline already fixed in MDB Part-4 Section 36 and Appendix-L Section 14 — an API release is not treated as exempt from that scheduling discipline merely because it may appear lower-risk than a database migration.

39. API Governance Standards

Fixes ownership and decision authority for the standards fixed across all four ASD Parts.

- The API Architecture Team is the designated owner of this ASD and all four of its Parts — any deviation from a standard fixed here requires that team's explicit, documented approval, not a unilateral per-module decision.
- This ASD is reviewed for continued accuracy at the same cadence as the other MDB/DDD documents it depends on (MDB Part-3 Section 30, DDD Part-3 Section 30) — a standard that has silently drifted from actual implementation practice is treated as a governance failure to be corrected, not tolerated as de facto reality.
- New API Groups introduced for a future module (per the extensibility pattern already fixed in Appendix-H Section 18.3 and MDB Part-3 Section 29) are held to the identical standards fixed across ASD Parts 1-4 — no future module receives a relaxed API standard merely for being newly introduced.
- Any conflict between this ASD and an earlier document in the chain (RGD v2.0 through DDD Part-4) is resolved in favor of the earlier, more foundational document — this ASD elaborates and applies those decisions, it does not have authority to silently override them.

40. API Final Review Checklist

The final consolidation point of the entire API Specification Document (Parts 1-4), verified before any major release, alongside the Production Readiness Checklist (MDB Part-4 Section 40) and Database Operations Checklist (DDD Part-4 Section 40).

#	Checklist Item
1	Every API Group's lifecycle stage (Section 31) is current and accurately recorded
2	No Deprecated version (Section 32) has exceeded its notice period without a retirement decision
3	Documentation artifact (Section 33) is regenerated and current for this release
4	Monitoring signals (Section 34) are confirmed flowing for every API Group in this release
5	Performance targets (Section 35) verified in Staging against production-representative load
6	Contract, Integration, and Regression Testing (Section 36) passed, including Backward Compatibility verification
7	Every breaking change (Section 37) carries API Architecture Team sign-off
8	Frontend/backend release coordination (Section 38) confirmed for any breaking change in this release
9	No undocumented deviation from ASD Parts 1-4 exists without a logged governance exception (Section 39)
10	ASD Part-2 (Section 20) and ASD Part-3 (Section 30) checklists both independently passed

A negative finding against any item above is treated as release-blocking, consistent with the Quality Gate discipline already fixed in Appendix-M Section 25 — this checklist is the API-specific instantiation of that same governing principle, completing the documentation chain from RGD v2.0 through to the operational API itself.